

Sample Space :

$$\text{Toss a coin} \rightarrow \Omega = \{HH, HT, TH, TT\}$$
$$P \rightarrow p_1, p_2, p_3, p_4$$

$$\sum p_i = 1$$

$$p_i \geq 0$$

event

$$A = \{ \text{At least 1 T appears} \}$$
$$= \{ HT, TH, TT \}$$

$$P(A) = p_2 + p_3 + p_4 = 1 - p_1$$
$$= 1 - P(A')$$

$$\mathcal{F} = \{ \text{subsets of } \Omega \} \rightarrow \underline{\text{ } \sigma\text{-field}}$$

A random variable X is a function $f: \Omega \rightarrow (-\infty, \infty)$

$X = \# \text{ of heads}$

$$X(\omega_1) = 2$$

$$X(\omega_2) = 1$$

$$X(\omega_3) = 1$$

$$X(\omega_4) = 0$$

$$X_2 = 5X_1$$

$$X_3 = \#H'_A - \#T'_A$$

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$X \sim \text{Bin}(n, p)$, $X = \text{H's in } n \text{ tosses}$ where $p = P(H)$

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k} \quad \text{for } k=0, \dots, n$$

$$X \sim \text{Bernoulli}(p) \Leftrightarrow X \sim \text{Binomial}(1, p)$$

$Y = \text{longest run of H's}$

$X_1, X_2, \dots$ on $\Omega = \{X_i : i \in \mathbb{N}\} \rightarrow \text{discrete}$

$\{X_t : t \geq 0\} \rightarrow \text{continuous}$
Stochastic Process $\rightarrow$ Sequence of these random variables

$F(x) = P(X \leq x) \rightarrow \text{Cumulative Distribution Function CDF}$

X_1, X_2 are independent if $P(X_1 \leq x_1, X_2 \leq x_2) = P(X_1 \leq x_1) \cdot P(X_2 \leq x_2)$

$X_1, X_2, \dots, X_n$ are mutually independent if

$$P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n) = P(X_1 \leq x_1) \cdot P(X_2 \leq x_2) \dots P(X_n \leq x_n)$$

for all $x_1, x_2, \dots, x_n \in (-\infty, \infty)$

(then ~~also~~ conversely true)

If X_i and X_j are independent for all i and j

Throw a fair dice twice.

X_1 = Sum of the outcomes is 7

X_2 = ~~the first throw of the dice is~~

X_3 = ~~the first throw of the dice is~~ " " " record "

$$\Omega = \{(i, j) : 1 \leq i, j \leq 6\}$$

$$P(X_2 = i, X_3 = j) = P(X_2 = i) \cdot P(X_3 = j)$$

$\nearrow 1/36$ $\downarrow 1/6$ $\downarrow 1/6$

X_2 and X_3 are independent

~~scribbled out text~~

$$P(X_1 = j, X_2 = k) = \begin{cases} 0 & \text{otherwise} \\ \frac{1}{36} & X_3 = j - k \end{cases}$$

$$P(X_1 = 7, X_2 = 3, X_3 = 4) = \frac{1}{36}$$

$$\neq \underbrace{P(X_1 = 7)}_{\frac{1}{6}} \cdot \underbrace{P(X_2 = 3) \cdot P(X_3 = 4)}_{\frac{1}{36}}$$

Defn $X_1, X_2, \dots$ is said to be independent sequence of r.v.'s if $P(X_1 \leq x_1, \dots, X_n \leq x_n) = \prod_{i=1}^n P(X_i \leq x_i)$ for all $-\infty \leq x_1, x_2, \dots, x_n \leq \infty$ and for all $n \geq 1$

Defn $X_1, X_2, \dots$ is said to be an independent and identically distributed sequence (iid) of r.v.'s if

(i) $X_1, X_2, \dots$ is an indep. seq. of r.v.'s

(ii) for all i, j $P(X_i \leq x) = P(X_j \leq x)$ for all $x \in \mathbb{R}$

HW: X, Y, Z mutually independent r.v's with ~~also~~ also identically distributed with $P(X=1) = \frac{1}{2}$, $P(X=-1) = \frac{1}{2}$. Consider the r.v's XY, YZ, XZ .

Q. Are they ^{mutually} independent? _/_/_

For a discrete r.v X : $E(X) = \sum x_i P(X=x_i)$

where $x_1, x_2, \dots$ are the values taken by X

$$\text{Var}(X) = E((X - E(X))^2) = E(X^2) - E(X)^2$$

$$\begin{aligned} \text{For r.v's } X, Y: \text{Cov}(X, Y) &= E((X - E(X))(Y - E(Y))) \\ &= E(XY) - E(Y)E(X) \end{aligned}$$

HW: 2) $X_1, X_2, \dots, X_{n-1}$ iid Bernoulli (p)

$$X_n = \begin{cases} 1 & \text{if } \sum_{i=1}^{n-1} X_i \text{ is odd} \\ 0 & \text{if } \sum_{i=1}^{n-1} X_i \text{ is even} \end{cases}$$

Q. Is the sequence $X_1, \dots, X_n$ indep? ~~Yes~~

Is the sequence $X_2, \dots, X_n$ indep? ~~Yes~~

Is the sequence $X_1, X_3, \dots, X_n$ indep?

$\{a_0, a_1, \dots\}$ finite or infinite

The generating function of this sequence is defined as

$$A(s) = \sum_{i=0}^{\infty} a_i s^i = \lim_{n \rightarrow \infty} \underbrace{\sum_{i=0}^n a_i s^i}_{\text{polynomial of } s \text{ in degree } n}$$

for $|s| < R$ where

R is Radius of convergence

$$\{1, 1, 1, \dots\} \rightarrow A(x) = \sum_{i=0}^{\infty} 1 \cdot x^i$$

$$= 1 + x + x^2 + x^3 + \dots$$

$$= \frac{1}{1-x} \quad \text{for } |x| < 1$$

↓
Radius of convergence

X is a non negative integer valued r.v

$$X = k, \quad k \geq 0$$

$$P(X=k) = p_k \Rightarrow \left(\sum_{k=0}^{\infty} p_k = 1 \right)$$



$$P(s) = \sum_{k=0}^{\infty} p_k s^k \rightarrow \text{Probability Generating Function}$$

Example: 1) $X \sim \text{Bernoulli}(p)$

$$X = \begin{cases} 1 & \text{w.p. } p \\ 0 & \text{w.p. } 1-p \end{cases}$$

$$P_X(s) = (1-p)s^0 + ps^1$$

2) $X \sim \text{Binomial}(n, p)$

$$P[X=k] = {}^nC_k p^k (1-p)^{n-k} \text{ for } k \in [0, n]$$

$$P_X(s) = \sum_{k=0}^n {}^nC_k p^k (1-p)^{n-k} s^k = \sum_{k=0}^n {}^nC_k (ps)^k (1-p)^{n-k}$$

$$\text{Binomial Theorem: } (a+b)^n = \sum_{k=0}^n {}^nC_k a^k b^{n-k}$$

$$\Rightarrow P_X(s) = (ps + 1-p)^n$$

→ Same as Bernoulli with $n=1$

3) $X \sim \text{Geometric}(p)$

X : Number of Failures till first Success
 $X \in [0, \infty)$

~~1~~

$$P(X=k) = (1-p)^k p \text{ for } k=0, 1, \dots$$

$$P_X(s) = \sum_{k=0}^{\infty} (1-p)^k p s^k = p \left(\sum_{k=0}^{\infty} ((1-p)s)^k \right)$$

$$= \frac{p}{1-s(1-p)}$$

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In general: $P_X(s) = \sum_{k=0}^{\infty} p_k s^k$

 for all $|s| \leq 1$

→ how!

$$P_X(1) = 1$$

6) $X \sim \text{Poisson}(\lambda)$

$$P[X=k] = \frac{e^{-\lambda} \lambda^k}{k!} \text{ for } k \geq 0$$

$$e^x = \underbrace{x^0}_{1!} + \underbrace{x^1}_{1!} + \underbrace{x^2}_{2!} + \underbrace{x^3}_{3!} + \dots$$

$$P_X(s) = \sum_{k=0}^{\infty} \frac{e^{-\lambda} \lambda^k}{k!} s^k = \sum_{k=0}^{\infty} \frac{e^{-\lambda} (\lambda s)^k}{k!}$$

$$= (e^{-\lambda}) (e^{\lambda s})$$

$$P_X(s) = e^{-\lambda(1-s)}$$

$$E(s^X) = \sum_{k=0}^{\infty} P[X=k] s^k$$

for a r.v X non neg integer valued

$$P_X(s) = \sum_{k=0}^{\infty} p_k s^k \quad |s| \leq 1$$

$$= p_0 s^0 + p_1 s^1 + p_2 s^2 + \dots$$

$$\frac{dP_X(s)}{ds} = 0 + p_1 + 2p_2 s + 3p_3 s^2 + \dots$$

$$\left. \frac{d}{ds} P_X(s) \right|_{s=0} = \mu_1 = P[X=1]$$

$$\left. \frac{d^2}{ds^2} P_X(s) \right|_{s=0} = \mu_2 = [2 P[X=2]]$$

$$\boxed{\left. \frac{d^n}{ds^n} P_X(s) \right|_{s=0} = [n P[X=n]]}$$

$$a = \{a_0, a_1, a_2, \dots\} \rightarrow A(s)$$

$$b = \{b_0, b_1, b_2, \dots\} \rightarrow B(s)$$

$$C_n = a_0 b_n + a_1 b_{n-1} + \dots + a_{n-1} b_1 + a_n b_0$$

$$= \sum_{i=0}^n a_i b_{n-i}$$

$$C = \{C_0, C_1, \dots\} \rightarrow C(s)$$

C is convolution of a and b

$$A(s)B(s) = (a_0 + a_1 s + a_2 s^2 + \dots)$$

$$\times (b_0 + b_1 s + b_2 s^2 + \dots)$$

$$= a_0 b_0 + s(a_1 b_0 + a_0 b_1) + s^2(a_2 b_0 + a_1 b_1 + a_0 b_2) + \dots$$

$\nearrow C_1 \qquad \qquad \qquad \nearrow C_2$

$$\Rightarrow A(s)B(s) = C(s)$$

Independent $\left\{ \begin{array}{l} X \text{ taking values } 0, 1, 2, \dots \\ Y \text{ taking values } 0, 1, 2, \dots \end{array} \right.$

$$\Rightarrow P[X=k, Y=L] = P[X=k] P[Y=L] \text{ for all } k, L$$

$$\begin{aligned}
 P(X+Y=j) &= P[X=0, Y=j] + P[X=1, Y=j-1] + \dots + P[X=j, Y=0] \\
 &= P(X=0)P(Y=j) + P(X=1)P(Y=j-1) + \dots + P(X=j)P(Y=0) \\
 &= p_0^x p_j^y + p_1^x p_{j-1}^y + \dots + p_j^x p_0^y
 \end{aligned}$$

$$P_{X+Y}(s) = P_X(s)P_Y(s) \text{ for } X, Y \text{ indep. r.v.s}$$

Notation: if the seq C is a convolution of the sequences a and b , then we write $C = a * b$

$$P_W(s) = P_Z(s) \iff W \text{ and } Z \text{ have same probability distribution, i.e. } W \text{ is same r.v. as } Z$$

(Proof by taking derivatives, and obtaining same probability at all points by setting $s=0$)

$$X_1, X_2, X_3, \dots, X_n \text{ iid Bernoulli}(p)$$

$$P_{X_k}(s) = (1-p) + ps = P_{X_k}(s), \quad k \in [1, n]$$

$$\begin{aligned}
 P_{X_1+X_2+X_3+\dots+X_n}(s) &= (1-p+ps)^n \\
 &= P_{\text{Binomial}(n,p)}(s)
 \end{aligned}$$

$$\Rightarrow (X_1 + X_2 + X_3 + \dots + X_n) \sim \text{Binomial}(n, p)$$

$$\left. \begin{array}{l} X_1 \sim \text{Poisson}(\lambda) \\ X_2 \sim \text{Poisson}(\mu) \end{array} \right\} \text{ indep}$$

$$\begin{aligned} P_{X_1+X_2}(s) &= P_{X_1}(s) \cdot P_{X_2}(s) \\ &= e^{-\lambda(1-s)} \cdot e^{-\mu(1-s)} = e^{-(\lambda+\mu)(1-s)} \\ &= P_{\text{Poisson}(\lambda+\mu)} \end{aligned}$$

$$\therefore (X_1 + X_2) \sim \text{Poisson}(\lambda + \mu)$$

$$E(X^k) = m_k \rightarrow k^{\text{th}} \text{ moment}$$

$$\frac{d^k}{ds^k} P_X(s) = \sum_{i=0}^{\infty} i(i-1)\dots(i-k+1) p_i s^{i-k}$$

$$\left. \frac{d}{ds} P_X(s) \right|_{s=1} = \sum_{i=0}^{\infty} i p_i = E(X)$$

$$\left. \frac{d^2}{ds^2} P_X(s) \right|_{s=1} = \sum_{i=1}^{\infty} i(i-1) p_i = E[X(X-1)]$$

$$\left. \frac{d^k}{ds^k} P_X(s) \right|_{s=1} = \underbrace{E[X(X-1)(X-2)\dots(X-k+1)]}_{k^{\text{th}} \text{ factorial moment of } X}$$

$$P_X'(1) = E[X]$$

$$P_X''(1) = E[X(X-1)] = E[X^2 - X] = E[X^2] - E[X]$$

$$\Rightarrow E[X^2] = P_X''(1) + P_X'(1)$$

If X and Y are 2 rv's $P(X=k|Y=L)$
 $= \frac{P(X=k, Y=L)}{P(Y=L)}$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\begin{aligned} \text{Var}[X] &= E[X^2] - (E[X])^2 \\ &= P_X''(1) - (P_X'(1))^2 \end{aligned}$$

drunk

A person takes a step in unit time

$$X_k = \begin{cases} 0 & \text{w.p. } \frac{1}{2} \\ 1 & \text{w.p. } \frac{1}{2} \end{cases}$$

X_k : Action at $t=k$

0 $\rightarrow$ stays, 1 $\rightarrow$ moves forward
 at time $= N$, he collapses
 No indep of steps

how far he has moved

$$S_N = \sum_{i=1}^N X_i$$

Theorem: Let $X_1, X_2, \dots$ be iid rv's and N a rv independent of $X_1, X_2, \dots$. Let $S_N = \sum_{i=1}^N X_i$

then $P_{S_N}(\lambda) = P_N(P_{X_1}(\lambda))$

Proof:
$$\begin{aligned} P_{S_N}(\lambda) &= \sum_{i=0}^{\infty} P(S_N=i) \lambda^i \\ &= \sum_{i=0}^{\infty} \lambda^i \sum_{j=0}^{\infty} P(S_N=i | N=j) P(N=j) \\ &= \sum_{i=0}^{\infty} \lambda^i \sum_{j=0}^{\infty} P(S_N=i, N=j) \end{aligned}$$

$$\sum_{j=0}^{\infty} P(S_N=i, N=j) = P(S_N=i)$$

$$\begin{aligned} &= \sum_{i=0}^{\infty} \lambda^i \sum_{j=0}^{\infty} P\left(\sum_{l=1}^j X_l = i \mid N=j\right) \\ &= \sum_{j=0}^{\infty} \sum_{i=0}^{\infty} \lambda^i P(N=j) P\left(\sum_{l=1}^j X_l = i\right) \end{aligned}$$

$$= \sum_{j=0}^{\infty} P(N=j) \sum_{i=0}^{\infty} s^i P\left(\sum_{l=0}^j X_l = i\right)$$

$$= \sum_{j=0}^{\infty} P(N=j) (P_{X_1}(s))^j \rightarrow t^j$$

$$= \sum_{j=0}^{\infty} P(N=j) t^j = P_N(t)$$

$$= P_N(P_X(s))$$

Example:

$X_1, X_2, \dots$ i.i.d.

$$X_1 = \begin{cases} 0 & \text{w.p. } \frac{1}{2} \\ 1 & \text{w.p. } \frac{1}{2} \end{cases}$$

$$P_{X_1}(s) = \frac{1}{2} + \frac{1}{2}s$$

$N \sim \text{geometric}\left(\frac{2}{3}\right)$

$$P_N(s) = \frac{2}{3-s}$$

$$P_{SN}(s) = P_N(P_{X_1}(s))$$

$$= P_N\left(\frac{1+s}{2}\right) = \frac{2}{3 - \left(\frac{1+s}{2}\right)} = \frac{4}{5-s}$$

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$$\text{pgf} : P_X(s) = E(s^X)$$

$$\text{mgf} : M_X(s) = E(e^{sx}) = \sum_{i=0}^{\infty} e^{si} P[X=i]$$

$$\left. \frac{d}{ds} M_X(s) \right|_{s=0} = \sum_{i=0}^{\infty} i P[X=i] = E[X]$$

$$\left. \frac{d^2}{ds^2} M_X(s) \right|_{s=0} = \sum_{i=0}^{\infty} i^2 P[X=i] = E[X^2]$$

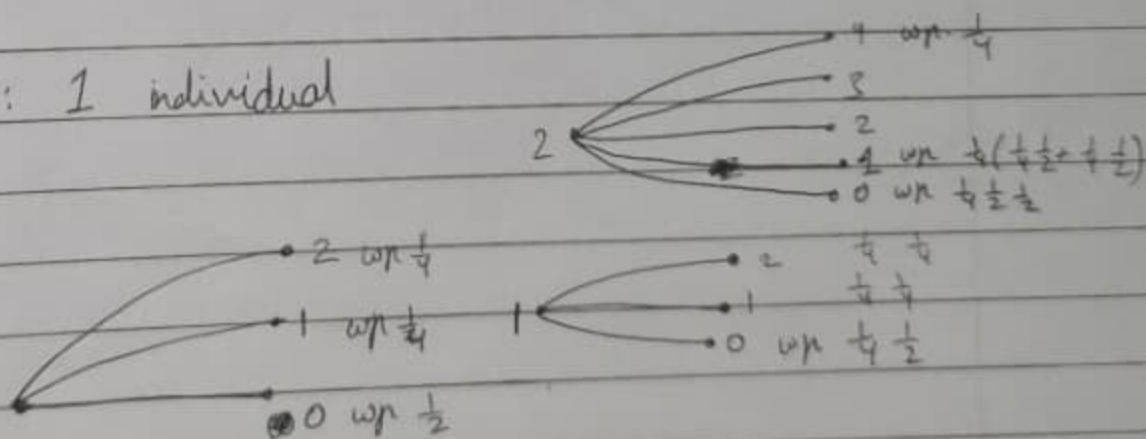
$$\left. \frac{d^k}{ds^k} M_X(s) \right|_{s=0} = E[X^k]$$

BRANCHING PROCESS
galton-watson process

of children of an individual is

0	w.p. $\frac{1}{2}$
1	w.p. $\frac{1}{4}$
2	w.p. $\frac{1}{4}$

gen 0 : 1 individual



H.W. 1

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1) Let X, Y, Z be 3 mutually independent random variables and assume that they are identically distributed with $P(X=1) = \frac{1}{2}$, $P(X=-1) = \frac{1}{2}$. Consider the random variables XY, YZ, ZX . Are they independent?

	Case (X, Y, Z)	Probability of this case	XY	YZ	ZX
1	$(1, 1, 1)$	$P(X=1, Y=1, Z=1) = \frac{1}{8}$	1	1	1
2	$(1, 1, -1)$	$\frac{1}{8}$	1	-1	-1
3	$(1, -1, 1)$	$\frac{1}{8}$	-1	-1	1
4	$(1, -1, -1)$	$\frac{1}{8}$	-1	1	-1
5	$(-1, 1, 1)$	$\frac{1}{8}$	-1	1	-1
6	$(-1, 1, -1)$	$\frac{1}{8}$	-1	-1	1
7	$(-1, -1, 1)$	$\frac{1}{8}$	1	-1	-1
8	$(-1, -1, -1)$	$\frac{1}{8}$	1	1	1

For mutual independence, $P(XY=1, YZ=1, ZX=1) = P(X=1, Y=1, Z=1) + P(X=-1, Y=-1, Z=-1) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$
 $P(XY=1) \cdot P(YZ=1) \cdot P(ZX=1) = (\frac{1}{2}) (\frac{1}{2}) (\frac{1}{2}) = \frac{1}{8}$
 $\therefore XY, YZ$ and ZX are not mutually independent

For pairwise independence:

Case 1: $XY=1, YZ=1$: $P(XY=1, YZ=1) = P(X=1, Y=1, Z=1) + P(X=-1, Y=-1, Z=-1)$
 $= 2(\frac{1}{8}) = \frac{1}{4}$

$$P(XY=1) P(YZ=1) = 4(\frac{1}{8}) \cdot 4(\frac{1}{8}) = \frac{1}{4}$$

Case 2: $XY=1, YZ=-1$: $P(XY=1, YZ=-1) = P(X=1, Y=1, Z=-1) + P(X=-1, Y=-1, Z=1)$
 $= 2(\frac{1}{8}) = \frac{1}{4}$

$$P(XY=1) P(YZ=-1) = 4(\frac{1}{8}) \cdot 4(\frac{1}{8}) = \frac{1}{4}$$

Case 3: $XY=-1, YZ=-1$: $P(XY=-1, YZ=-1) = P(X=1, Y=-1, Z=-1) + P(X=-1, Y=1, Z=1)$
 $= 2(\frac{1}{8}) = \frac{1}{4}$

$$P(XY=-1) P(YZ=-1) = 4(\frac{1}{8}) \cdot 4(\frac{1}{8}) = \frac{1}{4}$$

Case 4: $XY=-1, YZ=1 : P(XY=-1, YZ=1) = P(X=1, Y=-1, Z=-1) + P(X=-1, Y=1, Z=1)$
 $= 2\left(\frac{1}{8}\right) = \frac{1}{4}$

$$P(XY=-1)P(YZ=1) = 4\left(\frac{1}{8}\right) \cdot 4\left(\frac{1}{8}\right) = \frac{1}{4}$$

$\therefore$ By Symmetry, XY, YZ and ZX are pairwise independent.

2.) Let $X_1, X_2, \dots, X_n$ be i.i.d Bernoulli (p) random variables and

$$X_n = \begin{cases} 1 & \text{if } \sum_{i=1}^{n-1} X_i \text{ is odd} \\ 0 & \text{if } \sum_{i=1}^{n-1} X_i \text{ is even} \end{cases}$$

Is the sequence independent? Justify.

Ans. Without Loss of Generality, let $n=3$, so X_1, X_2 are iid Bernoulli(p) and $X_3 = \begin{cases} 1 & \text{if } X_1+X_2 \text{ is odd } (X_1=1, X_2=0 \text{ or } X_1=0, X_2=1) \\ 0 & \text{if } X_1+X_2 \text{ is even } (X_1=0, X_2=0 \text{ or } X_1=1, X_2=1) \end{cases}$

$$P(X_1=0, X_2=0, X_3=1) = 0 \quad \left(\begin{array}{l} \text{As when } X_1=0, X_2=0, \text{ then } X_3=0 \\ \text{as } X_1+X_2=0 \text{ is even} \end{array} \right)$$

$$\begin{aligned} P(X_1=0) \cdot P(X_2=0) \cdot P(X_3=1) &= (1-p)(1-p) \left(P[X_1=1, X_2=0] + P[X_1=0, X_2=1] \right) \\ &= (1-p)^2 (p(1-p) + p(1-p)) \\ &= 2p(1-p)^3 \end{aligned}$$

$\therefore P(X_1=0, X_2=0, X_3=1) = P(X_1=0) \cdot P(X_2=0) \cdot P(X_3=1)$ only if $p=0$ or $p=1$ (not for any $p \in [0, 1]$)

$\therefore X_1, X_2$ and X_3 are mutually dependent.

H.W. 2

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- 1) Let X be a non-negative integer valued random variable with probability generating function $P_X(s)$. Find the p.g.f of the random variables: (a) $Y = X+1$, (b) $Z = 2X$

Ans. In general, $P_X(s) = E[s^X]$

$$\begin{aligned} \text{(a)} \quad P_Y(s) &= E[s^Y] = E[s^{X+1}] \\ &= E[s^X \cdot s] \\ &= s E[s^X] \end{aligned}$$

$$\Rightarrow P_Y(s) = s P_X(s)$$

$$\begin{aligned} \text{(b)} \quad P_Z(s) &= E[s^Z] = E[s^{2X}] \\ &= E[(s^2)^X] \end{aligned}$$

$$\Rightarrow P_Z(s) = P_X(s^2)$$

In General for ~~any~~ X with p.g.f. $P_X(s)$ and $Y = aX + b$ where a, b are scalars, $P_Y(s) = s^b P_X(s^a)$

- 2) Let X be a non-negative integer valued random variable with probability generating function $P_X(s)$. For $n \geq 0$, let $a_n = P(X \leq n)$. Show that the generating function $A(s)$ of the sequence $\{a_n : n \geq 0\}$ is given by $A(s) = \frac{P_X(s)}{1-s}$.

Ans. For $n \geq 0$, $a_n = P(X \leq n)$. Let $A(s)$ be generating function of $\{a_n : n \geq 0\}$

$$A(s) = \sum_{i=0}^{\infty} a_i s^i = \sum_{i=0}^{\infty} P(X \leq i) s^i$$

$$= \sum_{i=0}^{\infty} s^i \left(\sum_{k=0}^i P(X \leq k) \right)$$

$$= s^0 (P(X \leq 0)) + s^1 (P(X \leq 0) + P(X \leq 1)) + s^2 (P(X \leq 0) + P(X \leq 1) + P(X \leq 2)) + \dots$$

$$= P(X \leq 0) (s^0 + s^1 + s^2 + \dots) + P(X \leq 1) (s^1 + s^2 + \dots) + \dots$$

$$= \sum_{k=0}^{\infty} P(X=k) \sum_{i=k}^{\infty} s^i$$

$$= \left(\frac{1}{1-\lambda} \right) \sum_{k=0}^{\infty} P(X=k) \cdot \lambda^k$$

$$= \frac{P_X(\lambda)}{1-\lambda}$$

3) Let X be a non negative integer valued random variable with probability generating function: $P_X(s) = 0.16 + p^2 s + p s^2 + 0.6 s^3$

a) Find the value of p

b) What is $P(X \geq 1)$

c) What is $E(X)$

Ans. a) ~~$P_X(1) = 1$~~ $P_X(1) = 1$

$$\Rightarrow 0.16 + p^2 + p + 0.6 = 1$$

$$\Rightarrow p^2 + p - 0.24 = 0$$

$$\Rightarrow p = \frac{-1 \pm \sqrt{1 + 4(0.24)}}{2}$$

$$\Rightarrow p = \frac{-1 \pm \sqrt{1.96}}{2} = \frac{-1 \pm 1.4}{2}$$

$$= 0.2, -1.2$$

Ans $P(X=2) = \frac{1}{2!} \frac{d^2}{ds^2} P_X(s) \Big|_{s=0} = \frac{1}{2!} 2p$

$$\Rightarrow \underline{p=0.2} \text{ as } P(X=2)=p \text{ is non negative.}$$

b) $P(X \geq 1) = 1 - P(X=0) = 1 - P_X(0) = 1 - 0.16$
 $= 0.84$

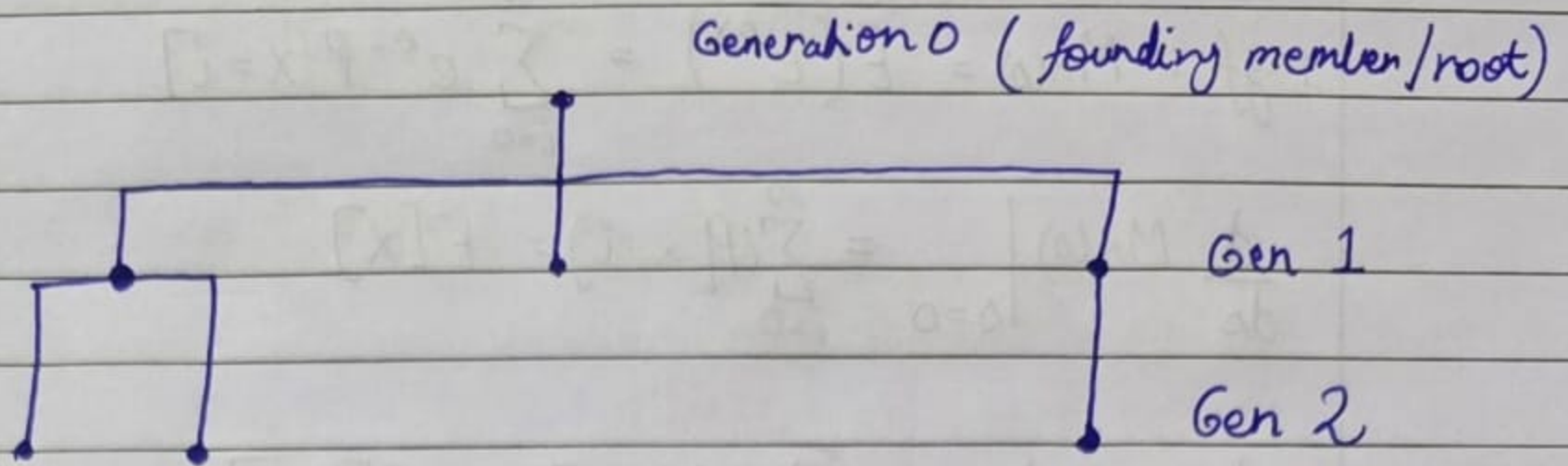
c) $E(X) = P'_X(1)$

$$= p^2 + 2ps + 3(0.6)s^2 \Big|_{s=1}$$

$$= p^2 + 2p + 1.8 = 0.04 + 0.4 + 1.8$$

$$= 2.24$$

Branching Processes



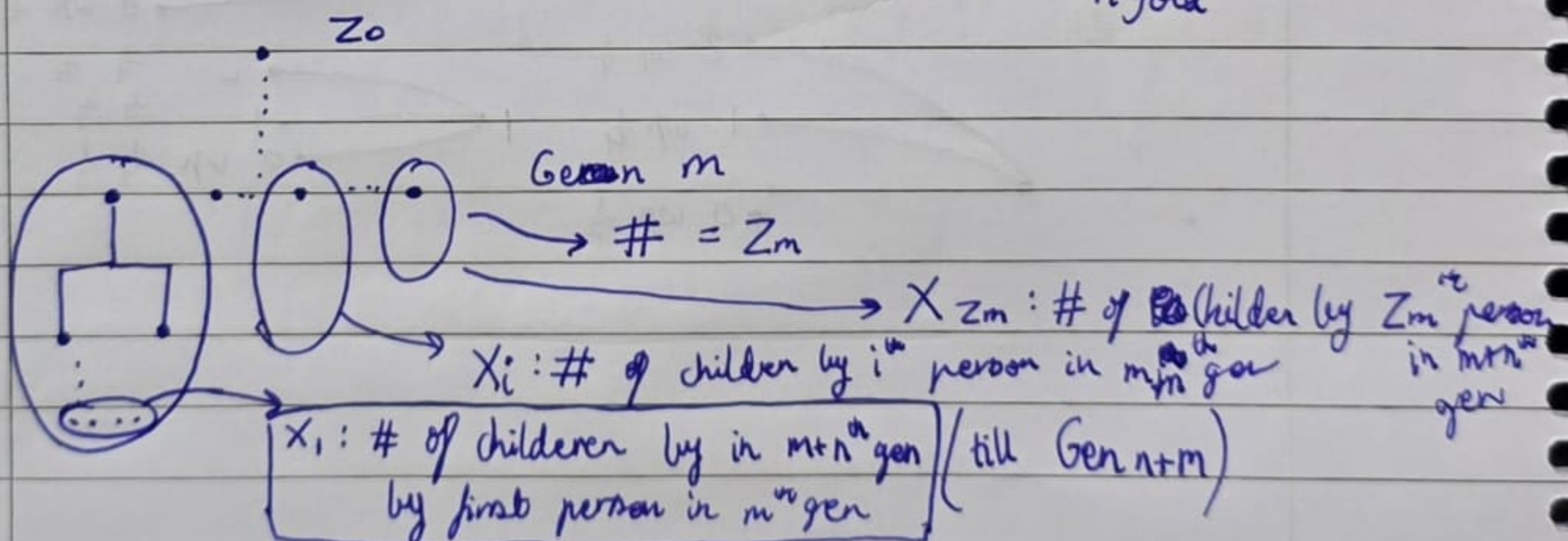
Assumptions:

- 1) Two distinct individuals produce children: the no. of children produced by each of the individuals are independent of each other. (mut or pairwise)
- 2) The no. of children produced by an individual is random with the same distribution as the no. of children produced by mother individual.

Notation: let $Z_n = \#$ of individuals in the n^{th} generation
 $Z_0 = 1$

Let $G_n(s) = P_{Z_n}(s)$ be the p.g.f. of Z_n

Theorem: $G_{n+m}(s) = G_n(G_m(s))$
 where $G_n(s) = \underbrace{G(G(G \dots (G(s))))}_{n \text{ fold}}$



i.i.d.

$X_1, X_2, \dots, X_{Z_m}$ are r.v.s with X_i being the # individuals in the $(n+m)^{\text{th}}$ generation have an ancestor i in the m^{th} gen.

$$Z_{m+n} = X_1 + X_2 + \dots + X_{Z_m}$$

$$\begin{aligned} G_{m+n}(s) &= P_{Z_{m+n}}(s) = \cancel{G_{m+n}(s)} \\ &= P_{Z_m}(P_{X_1}(s)) \\ &= \cancel{G_m}(G_n(s)) \end{aligned}$$

$$\begin{aligned} G(s) &= \text{p.g.f. of } Z_1 \\ G_n(s) &= \text{" " } Z_n \end{aligned}$$

→ Lemma: Let $\mu = E(Z_1)$, $\sigma^2 = \text{Var}(Z_1)$
 Then $E(Z_n) = \mu^n$ and $\text{Var}(Z_n) = \begin{cases} n\sigma^2 & \text{if } \mu=1 \\ \frac{\sigma^2(\mu^n-1)\mu^{n-1}}{\mu-1} & \text{if } \mu \neq 1 \end{cases}$

$$E[X] = \left. \frac{d}{ds} P_X(s) \right|_{s=1}, \quad \left. \frac{d^2}{ds^2} P_X(s) \right|_{s=1} = E[X(X-1)]$$

$$\begin{aligned} \frac{d}{ds} G_n(s) &= \frac{d}{ds} G(G_{n-1}(s)) \\ &= G'(G_{n-1}(s)) \cdot (G'_{n-1}(s)) \\ \therefore \left. \frac{d}{ds} G_n(s) \right|_{s=1} &= G'(G_{n-1}(1)) \cdot G'_{n-1}(1) \\ &= G'(1) \cdot E(Z_{n-1}) \end{aligned}$$

$$\Rightarrow E(Z_n) = n E(Z_{n-1}) \rightarrow \text{do induction}$$

$$\begin{aligned} \left. \frac{d^2}{ds^2} G_n(s) \right|_{s=1} &= \left. \frac{d^2}{ds^2} G(G_{n-1}(s)) \right|_{s=1} \\ &= \left. \frac{d}{ds} \left(G'(G_{n-1}(s)) \cdot G'_{n-1}(s) \right) \right|_{s=1} \\ &\vdots \\ &= G''(1) (G_{n-1}(1))^2 + G'(1) G''_{n-1}(1) \end{aligned}$$

Example: $Z_1 \sim \text{Geometric}(p)$

$$P(Z_1 = k) = (1-p)p^k, \quad k \geq 0$$

$$\text{let } 1-p=q$$

$$G(s) = P_{Z_1}(s) = \frac{q}{1-ps}$$

$$G_2(s) = G(G(s)) = \frac{q}{1-pG(s)} = \frac{q}{1-p\left(\frac{q}{1-ps}\right)} = \frac{q(1-ps)}{1-ps-pq}$$

$$G_n(s) = \begin{cases} \frac{n - (n-1)s}{n+1 - ns} & \text{if } p=q=\frac{1}{2} \\ \frac{q[p^n - q^n - ps(p^{n-1} - q^{n-1})]}{p^{n+1} - q^{n+1} - ps(p^n - q^n)} & \text{if } p \neq q \end{cases}$$

Use induction

Using $P(X=0) = P_X(0)$

—/—/—

$$P(Z_n=0) = \begin{cases} \frac{n}{n+1} & \text{if } p=q=\frac{1}{2} \\ \frac{q(p^n - q^n)}{p^{n+1} - q^{n+1}} & \text{if } p \neq q \end{cases}$$

$$\underbrace{\{Z_n=0\}}_{E_n} \subseteq \underbrace{\{Z_{n+1}=0\}}_{E_{n+1}}$$

$$P(E_n) \rightarrow P(\text{Ultimate extinction})$$

$$P(Z_n=0) \xrightarrow{n \rightarrow \infty} \begin{cases} 1 & \text{if } p \leq q \\ \frac{q}{p} & \text{if } p > q \end{cases} \quad \begin{matrix} p \leq \frac{1}{2} \\ p > \frac{1}{2} \end{matrix} \quad (p=1?)$$

$\rightarrow n \rightarrow \infty$ exists (limit exists)

Theorem: Suppose $P(Z_n=0) \rightarrow P(\text{ultimate extinction}) = \eta$
 η is the smallest non negative root of the equation
 $\lambda = G(\lambda)$.

Also $\eta = 1$ if $\mu < 1$

$\eta < 1$ if $\mu > 1$

$\eta = 1$ if $\mu = 1$ and $\sigma^2 > 0$

(connected to whether Z_1 is a const. r.v. or not)

Proof: Let $\eta_n = P(Z_n=0)$
 $\eta_n \uparrow$

$$\eta_n = G_n(0) = G(G_{n-1}(0)) = G(\eta_{n-1})$$

$$\eta = \lim_n \eta_n$$

η

$$\Rightarrow \eta = G(\eta)$$

(We need the fact that $G(\lambda)$ is continuous)

η is the smallest non-neg solⁿ

Let V be another solⁿ

$$\eta_1 = G(0) \leq G(V)$$

$$\eta_2 = G_2(0) = G(G(0))$$

~~$G(V)$~~

$$\rightarrow \eta_2 \leq G(G(V))$$

$$\eta_n \leq G_{n-1}(V)$$

$$\text{As } n \rightarrow \infty, \quad \eta_n \rightarrow \eta$$

$$G_{n-1}(V) \rightarrow V$$

$$\rightarrow \eta < V$$

$$P_X(s) = p_0 + p_1 s + \dots + p_k s^k$$

$$\leq P_X(t) \text{ for } s \leq t$$

$P_X(n)$ is a non decreasing function in s

Branching Processes

Z_n = # of individuals in the n^{th} gen

$$Z_0 = 1$$

Theorem: $P(Z_n = 0) \rightarrow P(\text{ultimate extinction}) = \eta$
 η is the smallest non-negative solⁿ of $s = G(s)$ $G(s)$ is the pgf of the offspring dist.

$$\eta = 1 \text{ if } \mu < 1$$

$$\eta > 0 \text{ if } \mu > 1$$

$$\eta = 1 \text{ if } \mu = 1 \text{ and } \sigma^2 > 0 \quad (\text{and } \eta = 0 \text{ if } \mu = 1 \text{ and } \sigma^2 = 0)$$

Proof: Let $\eta_n = P(Z_n = 0)$
 $\{Z_n = 0\} \subseteq \{Z_{n+1} = 0\}$
 η_n is non-decreasing, so $\eta_n \uparrow \eta$

$$\eta_1 = P(Z_1 = 0) = G(0) \leq G(1) \quad (\text{As generating func are non-decreasing})$$
$$\Rightarrow G(0) \leq 1$$

$$\eta_2 = G_2(0) = G(G(0)) \leq G(1)$$
$$\Rightarrow G(1) \leq 1$$

$$\eta_n = G_n(0)$$

$$= G(G_{n-1}(0)) \quad \downarrow \text{ By induction}$$
$$\leq G(1)$$

$$\Rightarrow \eta \leq 1$$

Observe $G'(s) = \frac{d}{ds} [p_0 + p_1 s + p_2 s^2 + \dots]$ $(p_i = P[Z_1 = i])$

$$= p_1 + 2p_2 s + 3p_3 s^2 + \dots$$

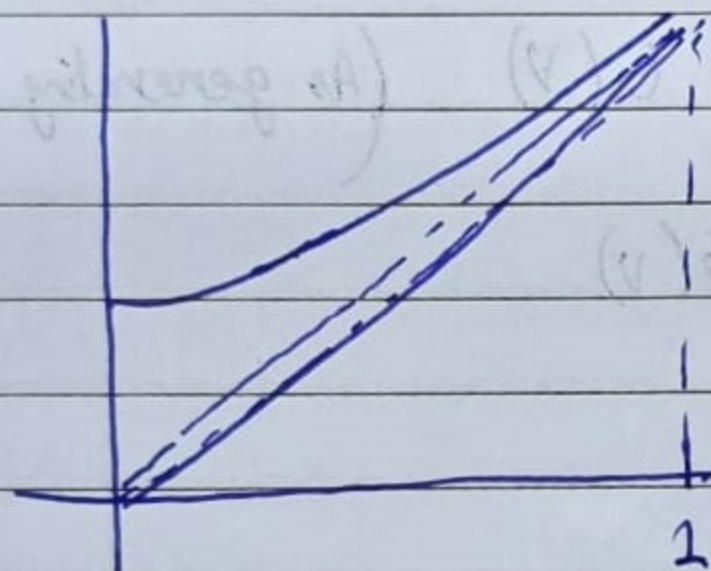
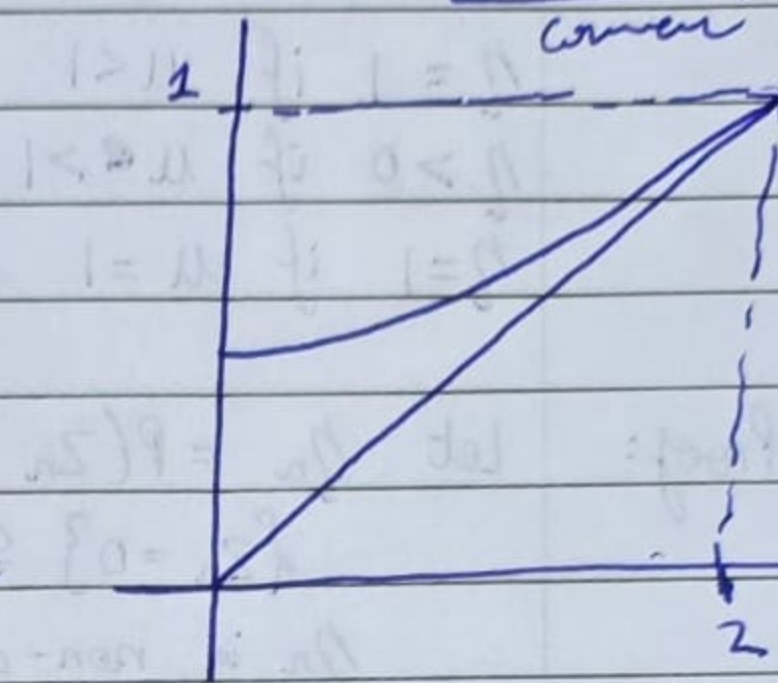
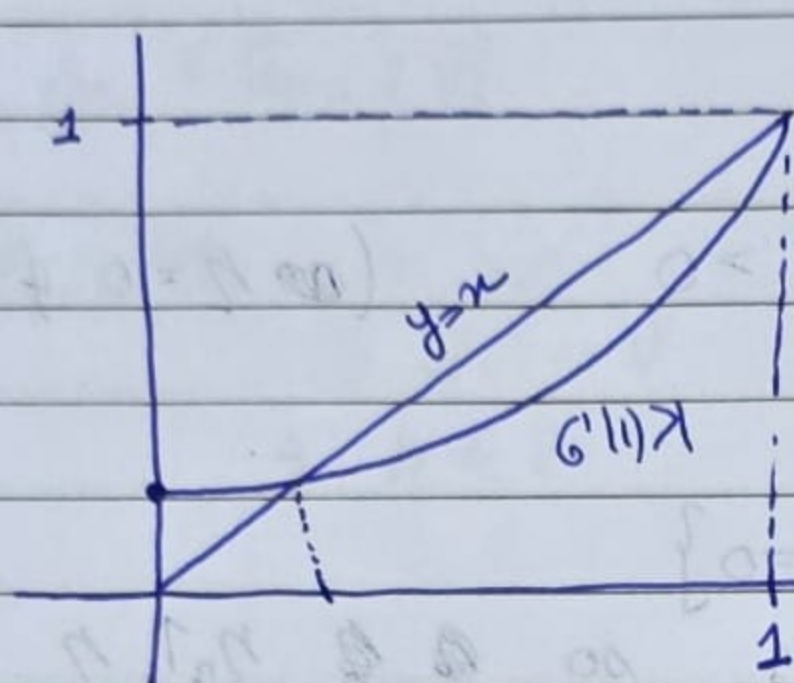
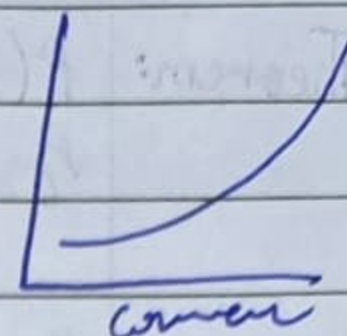
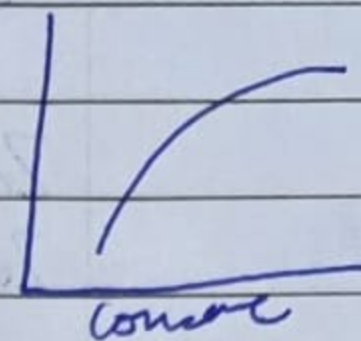
$$\geq 0$$

$$\therefore G'(s) \geq 0$$

$$G''(s) = \frac{d}{ds} (p_1 + 2p_2 s + 3p_3 s^2 + \dots)$$

$$= 2p_2 + 6p_3 s + \dots \geq 0 \quad (\text{for all } s \in [0, 1])$$

$\therefore G$ is a convex function



$$G(s) \leq \frac{1-s}{1-0} G(0) + \frac{s-0}{1-0} G(1) = (1-s)G(0) + sG(1)$$

$$G(1/2) \leq \frac{1-1/2}{1-0} G(0) + \frac{1/2-0}{1-0} G(1)$$

$$G(1/2) \leq \frac{1}{2} G(0) + \frac{1}{2} G(1)$$

$$2G(1/2) \leq G(0) + G(1)$$

$$0 \leq$$

$$G(1/2) \leq \frac{G(0) + G(1)}{2}$$

$X_1, X_2, \dots$ i.i.d.

$$X_1 = \begin{cases} +1 & \text{w.p. } p \\ -1 & \text{w.p. } q \end{cases} \quad \text{Rademacher}(p)$$

★ At even time, the summation is at even position and odd position at odd time

RW has walked till $2n$ times

$$P(S_{2n} = 2a) \text{ for } a \in \{-n, -n+1, \dots, n\}$$

$$\begin{aligned} r + l &= 2n \\ r - l &= 2a \end{aligned} \quad \Rightarrow \quad \begin{aligned} r &= n + a \\ l &= n - a \end{aligned}$$

$$\text{Then } P(S_{2n} = 2a) = {}^{2n}C_{n+a} p^{n+a} q^{n-a}$$

$$\boxed{\text{H.W. } P(S_{2n+1} = 2a+1) = ?}$$

Random Walk

$$X_1 \rightarrow P(X_1=1)=p, P(X_1=-1)=q=1-p$$

X_2

$\vdots$

X_n

$X_1, X_2, \dots, X_n$ are independent

$$S_N = \sum_{i=1}^N X_i, \quad S_0 = 0$$

$$= S_{N-1} + X_N$$

when $p=q$, it is called a simple symmetric

R.W.

(SSRW)

Peter and Paul Game:

A Rs. Peter

B Rs. Paul

gamble by toss a coin

if H then Paul gives 1 Rs. to Peter

if T then Peter gives 1 Rs. to Paul

Let T_n = fortune of Peter at time n

$$T_0 = A$$

$$T_1 = \begin{cases} A+1 & \text{w.p. } p \\ A-1 & \text{w.p. } 1-p \end{cases}$$

$$T_n = A + \sum_{i=1}^n X_i \quad \text{provided neither Peter or Paul have become bankrupt.}$$